

Filovirus pathogenesis and immune evasion: insights from Ebola virus and Marburg virus

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Abstract | Ebola viruses and Marburg viruses, members of the filovirus family, are zoonotic pathogens that cause severe disease in people, as highlighted by the latest Ebola virus epidemic in West Africa. Filovirus disease is characterized by uncontrolled virus replication and the activation of host responses that contribute to pathogenesis. Underlying these phenomena is the potent suppression of host innate antiviral responses, particularly the type I interferon response, by viral proteins, which allows high levels of viral replication. In this Review, we describe the mechanisms used by filoviruses to block host innate immunity and discuss the links between immune evasion and filovirus pathogenesis.

Negative-strand RNA viruses

Viruses possessing single-stranded RNA genomes that are negative sense (that is, complementary to the viral mRNAs).

Haemorrhagic fevers

Severe illnesses typically associated with bleeding that are caused by various viral infections, including filovirus infections.

In March 2014, cases of a severe, frequently lethal illness with symptoms that included fever, severe diarrhoea and vomiting were reported to be occurring in the West African country Guinea¹. The causative pathogen was determined to be Ebola virus (EBOV), a member of the filovirus family of enveloped, negative-strand RNA viruses noted for their ability to cause outbreaks of severe, often fatal, viral haemorrhagic fevers^{2,3}. Analysis of viral genome sequences and epidemiological investigation of the viruses from Guinea suggested an index case in a 2-year-old child in December 2013, with subsequent transmission of the virus to the prefectures of Guéckédou, Macenta and Kissidougou¹. The outbreak subsequently expanded into neighbouring Sierra Leone and Liberia. Consequent in-depth analysis of 78 EBOV sequences from Sierra Leone, obtained using deep sequencing approaches, helped to trace the spread of EBOV from Guinea and shed further light on transmission dynamics and genetic variation as the virus spread in the human population through mid-June 2014 (REF. 4). After a period of exponential spread in West Africa, the transmission of EBOV has greatly decreased, but has not ended. As of 16 September 2015, WHO had reported 28,220 confirmed, probable, and suspected cases of EBOV disease (EVD) in Guinea, Liberia and Sierra Leone, and 11,291 reported deaths. At least nine counties had cases although the vast majority were in West Africa⁵. This outbreak of EVD is unique in that it is by far the largest outbreak on record and the first outbreak of EVD in West Africa (BOX 1). The virus family *Filoviridae* is divided into three genera: Ebolavirus,

Marburgvirus and Cuevavirus⁶. Within the Ebolavirus genus are five species: Zaire ebolavirus (now known as EBOV), Sudan ebolavirus (SUDV), Bundibugyo ebolavirus (BDBV), Tai Forest ebolavirus (TAFV) and Reston ebolavirus (RESTV). The Marburgvirus genus consists of a single species, Marburg marburgvirus (MARV). The genome of another filovirus, Lloviu virus, which is the only member of the Cuevavirus genus, was demonstrated to be present in bats in northern Spain, but this virus has not successfully been cultured⁷. EBOV is responsible for the current outbreak in West Africa and along with SUDV, BDBV and MARV has caused past outbreaks of human disease; TAFV has only been associated with a single, non-fatal human infection (see [Outbreaks Chronology: Ebola Virus Disease](#) on the [CDC website](#)). RESTV appears to originate from the Philippines and has been isolated from non-human primates and pigs; infected non-human primates have been exported from the Philippines to other countries. Humans exposed to RESTV have not become ill, suggesting that RESTV is significantly attenuated in humans compared with other filoviruses⁸. Whether Lloviu virus is hazardous to humans is not yet known.

Filoviruses are suspected to use bats as reservoir hosts, although current evidence supporting this is stronger for MARV than for EBOV (BOX 2). As might be expected for a reservoir host, filoviruses do not seem to be overtly pathogenic in bats, despite the high virulence of these viruses in humans and non-human primates^{9,10}. The absence of fatal disease following filovirus infection is not a property unique to bats. For example, experimentally

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Box 1 | Filovirus outbreaks

The first documented outbreak caused by filoviruses occurred in 1967 in Marburg, Germany, when animal staff and laboratory workers were infected while harvesting tissues from *Cercopithecus aethiops* monkeys. These animals were imported from Uganda to produce kidney cell cultures needed for the production of a live poliomyelitis vaccine^{139,140}. In efforts to identify and characterize the infectious agent, plasma from infected guinea pigs was formalinized and analysed by electron microscopy. The causative agent was identified as a novel virus and named Marburg virus (MARV)¹⁴⁰. This pathogen is now classified in the Marburgvirus genus of the filovirus family. In 1976, another outbreak of haemorrhagic fever occurred in northern Zaire, now the Democratic Republic of the Congo (DRC), resulting in 318 cases with an 88% lethality rate. The causative agent of this outbreak was another filovirus, Ebola virus (EBOV)¹⁴¹. Ebolavirus represents the second genus of the filovirus family. In the same year, a second outbreak of haemorrhagic fever caused by filovirus also occurred in Sudan, causing 284 cases with a 53% lethality rate¹⁴². This second outbreak was caused by a distinct virus, Sudan ebolavirus (SUDV). In the years since these early outbreaks, small sporadic outbreaks of MARV infection have occurred in Africa, with the largest epidemics occurring in 1998 in the DRC, with 141 cases and an 82% lethality rate, and in 2004–2005 in Angola, with 252 cases and a 90% lethality rate^{143,144}. Subsequent outbreaks of EBOV revealed the presence of three additional species with varying pathogenicity in humans. In 1989, Reston ebolavirus (RESTV) was isolated from cynomolgus macaques in the Philippines and was found to be non-pathogenic in humans¹⁴⁵. In 1994, the Ivory Coast ebolavirus (now called Tai Forest ebolavirus (TAFV)) was discovered in an infected patient in the Tai Forest reserve in Cote d'Ivoire¹⁴⁶. Bundibugyo ebolavirus (BDBV) was discovered in 2007 in Uganda, where it caused 149 cases and a 25% lethality rate. Most human cases of EBOV disease (EVD) are due to the emergence and re-emergence of EBOV in rural areas of Gabon, the Republic of Congo and the DRC, and of SUDV in Sudan and Uganda. Until 2014, there had been 1,383 EBOV and 779 SUDV cases, with fatality rates ranging from 50% to 90% and from 40% to 65%, respectively (see Outbreaks Chronology: Ebola Virus Disease on the CDC website). The 2014 West Africa EBOV outbreak is unique in that it is the largest outbreak on record to date. The most recently identified filovirus, Lloviu virus (LLOV), which is classified in a distinct filovirus genus Cuevavirus, was identified in bats in Spain⁷. LLOV has yet to be cultured, but a near full-length viral genomic sequence has been determined from RNA in bat tissue. The pathogenic potential of LLOV is uncertain.

infected guinea pigs and mice do not die of filovirus infection, although serial passage of EBOV or MARV in these species can select for viruses that are lethal to these animals^{11–14}. Importantly, the factors that determine whether a filovirus is pathogenic in a particular host are not well defined, although the type I interferon (IFN) response has a role in containing infections in mice¹⁵ (see below).

Filoviruses possess single-stranded, negative-sense RNA genomes of approximately 19 kilobases. These genomes have seven distinct genes, each of which functions as a separate transcriptional unit (FIG. 1). These genes encode seven viral structural proteins known as nucleoprotein (NP), viral protein 35 (VP35), VP40, glycoprotein (GP), VP30, VP24 and the large protein (L) (FIG. 1a). For the ebolaviruses, the GP gene encodes multiple distinct proteins owing to transcriptional editing by the viral RNA-dependent RNA polymerase (RdRp), which adds non-template-encoded adenosine nucleotides to the nascent mRNA. The unedited transcript encodes the non-structural soluble GP (sGP), which is secreted, whereas editing results in translational frameshifting such that the membrane-bound GP and a second secreted GP are produced from edited RNAs^{3,16,17}. Additional editing events that have been

recently described for both EBOV and MARV mRNAs, in which non-template-encoded nucleotides are added at sites other than the canonical editing site in the GP mRNA, may further diversify the proteins translated from filovirus mRNAs¹⁸.

With the exception of EBOV sGP, each viral protein is present in the enveloped viral particle. GP is a type I transmembrane protein and is the only viral protein exposed on the surface of viral particles. GP mediates entry into host cells, acting as the attachment and fusion protein (FIG. 1b). Viral internalization occurs by macropinocytosis. Within the cytoplasm, viral RNA synthesis occurs on the uncapped, non-polyadenylated template viral RNA, which is associated with the NP. The NP, VP35, VP30 and L proteins are all required for viral RNA synthesis. The L protein is the enzymatic component of the RdRp complex and in addition to RdRp activity possesses methyltransferase and guanylttransferase activities, which are needed to cap the 5' end of viral mRNAs. VP35 acts as a viral polymerase cofactor: it interacts with both the L protein and NP to deliver the L protein to the NP-encapsidated template. VP30 is required for viral transcription (mRNA synthesis). The phosphorylation state of VP30 may also regulate the transcriptase and replicase functions of the viral polymerase complex. During viral replication, tubular nucleocapsids form in the cytoplasm of the infected cell, and transfection-based studies indicate that these consist of viral RNA, NP, VP35 and VP24 (FIG. 1b). The nucleocapsids are delivered to sites of viral assembly and are released at the plasma membrane. Virus release occurs by budding and involves the interaction of VP40 with components of the cellular endosomal sorting complex required for transport (ESCRT) pathway. VP40 is sufficient for viral budding, but budding efficiency is enhanced by other viral proteins, including GP, which reaches the plasma membrane (FIG. 1b).

Filovirus pathogenesis is best characterized for the EBOV species, particularly in non-human primates. Importantly, the severity of filovirus infection in humans and non-human primates seems to result from robust systemic virus replication and excessive inflammatory responses, which lead to the manifestations of filoviral disease, including high fever, vascular leakage and coagulopathy². Underlying the uncontrolled viral replication is the potent suppression of innate antiviral defences by filovirus gene products¹⁹, including EBOV VP35 (eVP35), MARV VP35 (mVP35), EBOV VP24 (eVP24) and MARV VP40 (mVP40), which disable specific innate immune signalling pathways, and the GP of either EBOV or MARV, which counteracts the antiviral effects of the host protein tetherin (also known as bone marrow stromal antigen 2 (BST2) and CD317)^{20–24}. In this Review, we describe the mechanisms of immune evasion that are mediated by filoviral proteins and discuss how these mechanisms are related to pathogenesis. Furthermore, we describe how filoviruses have evolved mechanisms to take advantage of host cellular pathways in order to increase the survival of infected cells and to promote optimal levels of viral protein translation.

Index case

The first identified case of a particular disease or epidemic, or the case from which the disease initially progressed.

Type I interferon

A group that comprises interferon- β and multiple types of interferon- α .

Macropinocytosis

An actin-dependent form of endocytosis that results in the formation of large fluid-filled vacuoles.

Box 2 | **Bats as filovirus reservoir hosts**

Elucidation of the reservoir host that harbours filoviruses has long been of interest. The identification of the reservoir might suggest mechanisms that drive the transmission of the viruses into human populations and strategies to avoid human infection. Because it is presumed that the reservoir host does not suffer the extreme mortality seen in humans and non-human primates, studies of virus–reservoir interaction could also provide unique insight into filovirus pathogenesis. However, it is only recently that conclusive evidence has been provided implicating select bat species as natural hosts of Marburg virus (MARV).

Circumstances surrounding several MARV outbreaks suggested bats as a possible source of the virus for human infection. For example, bats were one of several animals to which the presumed index case of a small MARV outbreak in 1975 had been exposed¹⁴⁷. In several cases, human infections occurred in individuals, including miners and tourists, who had entered caves containing significant bat populations^{144,148–150}. *Rousettus aegyptiacus* was repeatedly found in these caves¹⁵¹. More direct evidence of MARV infection of bats was the detection of MARV nucleic acids and MARV-specific antibodies in *R. aegyptiacus*¹⁵¹. However, the conclusive evidence was the isolation of live virus from *R. aegyptiacus*¹⁴⁹. Follow-up studies, which focused on Python Cave in Uganda, where two tourists had become infected previously, found evidence of a seasonal pattern of infection in the bat population¹⁵². Consistent with the expected outcome that infection of a reservoir host should not be consistently lethal, experimental inoculation of filoviruses into bats has not yielded signs of overt disease^{9,10}. Interestingly, current data suggest that MARV infections of *R. aegyptiacus* are not persistent but instead are cleared after a period of replication. The animals develop a transient viraemia but mount a detectable antibody response to the infection, and the virus is cleared. However, oral shedding, as well as replication, in multiple bat tissues has been detected. These data suggest potential routes that might lead to spread of infections among bats or to other species, including humans, although transmission between experimentally infected bats has not been demonstrated^{9,10}.

The evidence for bats as Ebola virus (EBOV) reservoirs is currently based on serology and the presence of EBOV-derived RNAs in tissues and sera from bats in both Africa and Asia⁹. The near complete genome of Lloviu virus (LLOV) was characterized from bats in northern Spain⁷. The isolation of infectious virus from bats, particularly from healthy bats, would provide stronger evidence that bats are reservoir hosts for EBOV and LLOV. Nonetheless, the available data suggest that filoviruses may have a broader geographical distribution than is generally appreciated.

With the finding that bats are susceptible to MARV infection, but resistant to disease, an understanding of the mechanisms of disease resistance is needed. It will be of significant interest to determine how interactions of filoviruses with the bat innate or adaptive immune responses somehow contribute to disease resistance.

Filovirus pathogenesis

Human infections caused by filoviruses are acquired by direct contact with infected bodily fluid, and the viruses are likely to enter the human body through breaks in the skin or mucosal surfaces²⁵. Studies using non-human primates, the gold standard animal model for studying EVD, have provided significant insight into the pathogenesis of filovirus infections. Immunohistochemistry and *in situ* hybridization demonstrate that filoviruses can infect many cell types, including macrophages, monocytes, dendritic cells, Kupffer cells in the liver, fibroblasts, hepatocytes, cells of adrenal gland tissue, endothelial cells and epithelial cells^{26,27}. Time-course studies of infection with EBOV show that the virus preferentially and initially replicates in macrophages and dendritic cells²⁷. These observations are consistent with histopathology studies carried out on human tissues²⁸. A definitive basis for the preferential infection of macrophages and dendritic cells is lacking. However, EBOV has been demonstrated to effectively infect macrophages and dendritic cells following their differentiation from monocytes^{29–31}. Differentiation into macrophages and dendritic cells

correlates with the upregulation of expression of cellular cathepsins and the protein Niemann–Pick C1 (NPC1), which are factors involved in GP-mediated membrane fusion^{31–35}. Further specificity for macrophages and dendritic cells may be promoted through the interaction of the virus with cell surface attachment receptors, including dendritic-cell-specific ICAM3-grabbing non-integrin (DC-SIGN; also known as CD209) and phosphatidylserine-binding proteins^{26,27,36–39}. Owing to the migratory characteristics of dendritic cells and macrophages, infection of these cells allows the transport of the virus by the lymphatic system to lymph nodes and through the blood to the liver and spleen, where the virus can infect resident tissue macrophages, including Kupffer cells, and resident dendritic cells^{26,27}. *In vivo*, filoviruses trigger the release of pro-inflammatory cytokines and chemokines, such as interleukin-1 β (IL-1 β), IL-6, IL-8, IL-10, monocyte chemoattractant protein 1 (MCP1; also known as CCL2), macrophage inflammatory protein 1 α (MIP1 α ; also known as CCL3), MIP1 β (also known as CCL4) and tumour necrosis factor (TNF), in addition to reactive oxygen species and nitric oxide^{40–45}. These factors are probably produced by infected macrophages^{42,46–48}. MIP1 α and MCP1 can recruit additional macrophages to areas of infection, enabling EBOV to infect more target cells²⁷. Infection of dendritic cells inhibits their maturation and thereby prevents antigen presentation to T cells. For example, *in vitro* studies have shown that EBOV infection inhibits upregulation of the co-stimulatory molecules CD40, CD80, CD86 and major histocompatibility complex (MHC) class II molecules^{49–51}. These dendritic cell maturation-suppressing functions have been linked to suppression of innate immune signalling pathways by viral proteins^{49,51–53}. However, the extent to which inhibition of dendritic cells influences adaptive immune responses remains to be determined, as EBOV-infected mice and people who survive infection have been demonstrated to develop an adaptive immune response to viral antigens^{54,55}.

In addition to interfering with the maturation and function of macrophages and dendritic cells, EBOV infection has consequences for B cell and T cell responses. For example, lymphopenia has been detected following infection with EBOV and MARV⁵⁶. For EBOV, this occurs most notably among CD4⁺ T cells, CD8⁺ T cells and natural killer (NK) cells, at 2 to 3 days post-infection in mouse models and at 4 days post-infection in non-human primates^{57,58}. Similarly, EBOV infection results in the loss of human CD4⁺ and CD8⁺ T cells *in vitro* at 4 days post-infection⁵⁹. Apoptosis is likely to contribute to the loss of lymphocytes, and several mechanisms have been proposed to explain this phenomenon^{42,45,56,60–64}. The consequences of lymphopenia for pathogenesis are not fully elucidated, and in an EBOV mouse model of infection prevention of lymphocyte apoptosis did not prevent death of infected mice⁶¹. However, loss of CD4⁺ T cells, which are required for isotype class switching in B cells, could contribute to the lack of EBOV-specific immunoglobulin M (IgM) and IgG antibodies in case fatalities⁶⁰. Another way in which EBOV may modulate B cell responses is through sGP, which may redirect GP-specific antibody responses to non-neutralizing epitopes⁶⁵.

Cathepsins

Endosomal cysteine proteases that cleave the filovirus glycoprotein during viral entry.

Phosphatidylserine

A phospholipid found on the surface of apoptotic bodies and on viral membranes that can promote intracellular uptake through interaction with specific cell surface receptors.

Lymphopenia

Low levels of lymphocytes in the blood.

Isotype class switching

A recombination process that alters the isotype of an antibody (for example, from immunoglobulin M to immunoglobulin G).

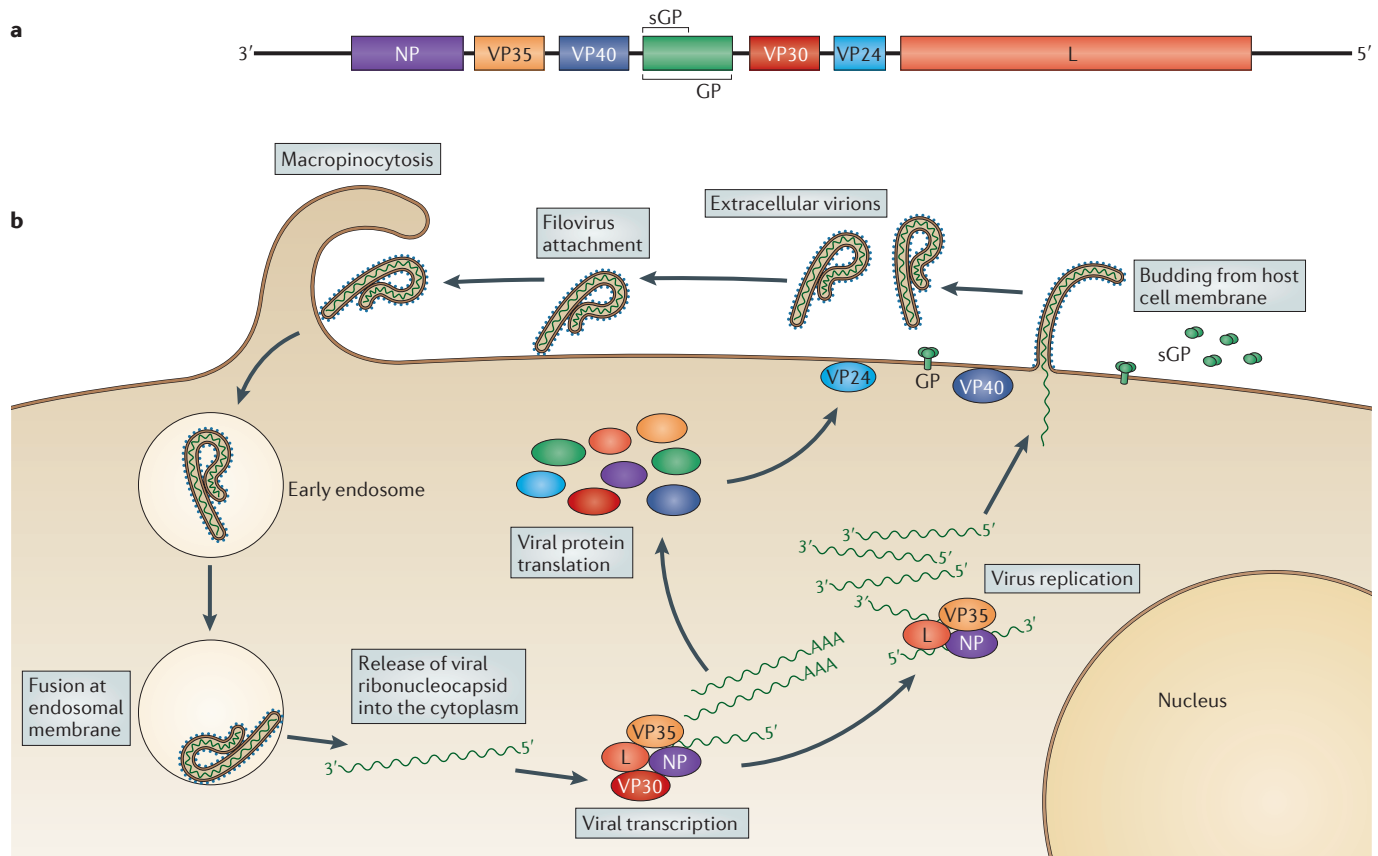


Figure 1 | Filovirus genome structure and life cycle. **a** | The genome of Ebola virus (EBOV) is depicted, in the 3'-to-5' orientation, to indicate that the genomic RNA is negative sense. The proteins encoded are nucleoprotein (NP), viral protein 35 (VP35), VP40, glycoprotein (GP), soluble GP (sGP), VP30, VP24 and large protein (L). Note that GP and sGP are encoded by a single gene in EBOV owing to transcriptional editing. Marburg virus (MARV) does not encode sGP. Filoviruses have unusually long non-coding regions at the 5' and 3' ends of their mRNAs. The genome regions corresponding to these non-protein-coding sequences are not drawn to scale. **b** | The general life cycle of a filovirus is displayed. First, the filovirus attaches to the cell membrane by its surface GP. The virus is then taken up by a process known as macropinocytosis^{37,153–155}. Upon acidification of the endosome, the cellular proteases cathepsin B and cathepsin L cleave GP³². This allows GP and the host protein Niemann–Pick C1 (NPC1) to interact, which is a prerequisite for the fusion of viral and endosomal membranes^{33,34}. Host endosomal calcium channels called two-pore channels play a crucial part in the endosomal trafficking of incoming viral particles to the site of fusion¹⁵⁶. GP mediates fusion of the viral and the endosomal membrane, releasing the viral ribonucleocapsid into the cytoplasm, where the negative-strand RNA genome undergoes transcription and replication¹⁵⁷. Production of 5'-capped, 3'-polyadenylated mRNAs from individual viral genes occurs, and genome replication follows, in which the genomic RNA template is copied into a full-length complement. The full-length complement serves as a template for the synthesis of additional negative-sense genomes. The RNA synthesis reactions require NP, VP35 and L. For initiation of transcription, EBOV also requires VP30. Translation of viral proteins occurs, and new viral particles are assembled at the plasma membrane. VP40 functions as the viral matrix protein and directs budding of particles from the cell surface^{158–160}. GP, a type I transmembrane protein, is incorporated into the budding particles, as are viral nucleocapsids containing the viral genome¹⁶¹.

Haemorrhage and intravascular coagulation are features of severe EBOV infection and may also be dependent on the ability of EBOV to suppress immune responses. For example, EBOV immune evasion may facilitate viral dissemination to hepatocytes, adrenal cortical cells and endothelial cells of connective tissue²⁷. Hepatocellular necrosis may result in the decreased synthesis of coagulation proteins, whereas adrenocortical infection and necrosis may negatively affect blood pressure homeostasis, potentially leading to haemorrhaging²⁷. The increased vascular permeability and leakage that typically lead to haemorrhagic fever could

be attributed to the release of high levels of TNF from infected macrophages, which can increase endothelial permeability⁴². Similarly, the release of nitric oxide, which is an important effector molecule in the homeostasis of the cardiovascular system, by infected macrophages may result in loss of vascular smooth-muscle tone and hypotension^{43,66}.

Studies in non-human primates have provided direct evidence that coagulation abnormalities are initiated during the early stage of EBOV infection. A dramatic decrease in plasma levels of anticoagulant protein C occurred early post-infection, after approximately

Necrosis

A non-apoptotic form of cell death.

2 days. This was followed by an increase in both the levels of tissue plasminogen activator, which is involved in dissolving blood clots, and of fibrin-degradation products (D-dimers) at 5 days post-infection⁶⁷. In addition, EBOV infection of macrophages leads to the upregulation of tissue factor on the surface of monocytes and macrophages and the release of membrane microparticles that express tissue factor. This increase in the level of tissue factor is likely to promote overactivation of the extrinsic pathway of coagulation and may cause disseminated intravascular coagulation⁶⁷. It is postulated that expression of tissue factor may be further upregulated by pro-inflammatory cytokines such as IL-6 that are abundant during acute EBOV infection, thereby exacerbating the intravascular coagulation phenotype⁶⁸. Thrombocytopenia and prolonged pro-thrombin time are also indicators of dysregulation of blood coagulation. Fibrinolysis during filovirus infection may manifest as petechiae, ecchymoses, mucosal haemorrhages and congestion^{67,69}. The compounding effects of an increased level of procoagulation factors and a decreased level of anticoagulant factor protein C could plausibly explain the activation of coagulation and haemorrhagic features that are characteristic of filovirus infection. Notably, towards the terminal stage of infection and after the onset of haemorrhagic abnormalities, EBOV replicates in endothelial cells²⁶. However, although endothelial cells are thought to have a role in pathogenesis, the molecular mechanisms that affect endothelial cell function are not yet fully understood^{26,70–73}.

In addition to haemorrhage and intravascular coagulation, the lethality of filovirus infection is probably due, in part, to the ability of filoviruses to replicate largely unimpeded by the host innate antiviral response. This innate immune evasion is likely to facilitate the excessive viral replication that activates the damaging host responses outlined above. Underlying this robust growth are multiple mechanisms that enable filoviruses to counteract the host innate antiviral defences, particularly IFN responses¹⁹. The type I IFNs (IFN β and multiple types of IFN α) constitute the major components of the innate immune response to viral infections⁷⁴. Expression of type I IFNs can be triggered by signalling from pattern recognition receptors, including RIG-I-like receptors (RLRs) such as RIG-I, melanoma differentiation-associated protein 5 (MDA5; also known as IFIH1) and LGP2 (also known as DHX58), and select Toll-like receptors (TLRs) such as TLR3, TLR4, TLR7 and TLR9. The RLRs are cytoplasmic RNA helicases that detect viral RNAs and promote IFN responses. Of these, RIG-I and MDA5 seem to be most relevant to filovirus infection (FIG. 2a). RIG-I and MDA5 are activated by ‘immunostimulatory RNAs’ such as RNAs with 5' triphosphates and double-stranded RNAs (dsRNAs) that may be produced during filovirus replication. Following their activation, RIG-I and MDA5 activate the serine/threonine kinases TBK1 and I κ B kinase- ϵ (IKK ϵ), which phosphorylate the transcription factors IFN regulatory factor 3 (IRF3) and IRF7. This promotes the nuclear accumulation of IRF3 and IRF7, which can engage IFN-stimulated response elements (ISREs),

thereby triggering the expression of type I IFNs (FIG. 2a). When expressed, type I IFNs are secreted from cells and signal through a heterodimeric receptor, the type I IFN receptor (IFNAR); this activates a JAK–STAT (Janus kinase–signal transducer and activator of transcription) signalling pathway and induces the expression of numerous IFN-stimulated genes (ISGs), triggering an antiviral state⁷⁵ that renders cells refractory to viral infection.

EBOV and MARV effectively counter this crucial antiviral defence system, blocking both the production of type I IFNs and the cellular responses to exogenously added type I IFNs^{24,46,76–78}. The importance of this inhibition is suggested by the observation that EBOV genomic RNA can activate RIG-I; that activation of RIG-I signalling before EBOV infection substantially reduces virus growth; and that recombinant EBOVs that are unable to block IFN responses are attenuated both in cell culture and in experimentally infected animals^{15,79–82}. The mechanisms by which type I IFNs may exert their anti-filoviral effects have not been systematically studied, but several proteins encoded by ISGs exert anti-filoviral effects in cell culture. For example, IFN-induced transmembrane (IFITM) proteins inhibit filovirus entry into host cells, and ISG15 is reported to impair viral budding mediated by VP40 (REFS 83,84). Nonetheless, although IFNs can control infection in rodent models, filoviral infections are not controlled by host defences in non-human primates and death of the host frequently ensues^{13,15,85,86}.

Subversion of cytosolic sensing

EBOV and MARV very effectively impair type I IFN production by interfering with the ability of the host cell to detect the presence of viral products in the cytosol. This subversion of cytosolic sensing of viruses is largely mediated by VP35 (FIG. 2a).

VP35 suppresses cytosolic sensing. VP35 is a multifunctional viral protein, serving not only as an antagonist of the innate immune response but also functioning as a non-enzymatic cofactor for the viral RdRp and as a viral structural protein. The VP35 proteins of EBOV and MARV block signalling through RLRs and prevent the phosphorylation of IRF3 and IRF7, thereby impairing the IFN response⁸⁷ (FIG. 2a). These inhibitory activities probably reflect inhibition at both upstream and downstream steps in the RLR signalling cascades.

Downstream targets of VP35 include the kinases IKK ϵ and TBK1. For example, VP35 has been demonstrated to interact with and act as a decoy substrate for these kinases such that IRF3 interaction with IKK ϵ and TBK1 is blocked and VP35 is phosphorylated instead⁸⁸. VP35 also interacts with IRF7, SUMO-conjugating enzyme UBC9 and SUMO E3 protein ligase PIAS1 to promote sumoylation of IRF7, a modification that impairs IRF7-dependent transcriptional responses^{89,90}.

Inhibitory activities upstream of RIG-I are connected to the dsRNA-binding capacity of VP35, which is mediated by residues within the carboxy-terminal IFN inhibitory domain (IID) of VP35⁹¹. Although wild-type

Fibrin

A non-globular protein involved in blood clotting.

Extrinsic pathway of coagulation

The blood coagulation pathway that is initiated at the site of injury in response to tissue factor.

Thrombocytopenia

Low levels of platelets in the blood.

Pro-thrombin time

A blood test that measures clotting time.

Sumoylation

A post-translational modification that influences protein stability and other cellular processes.

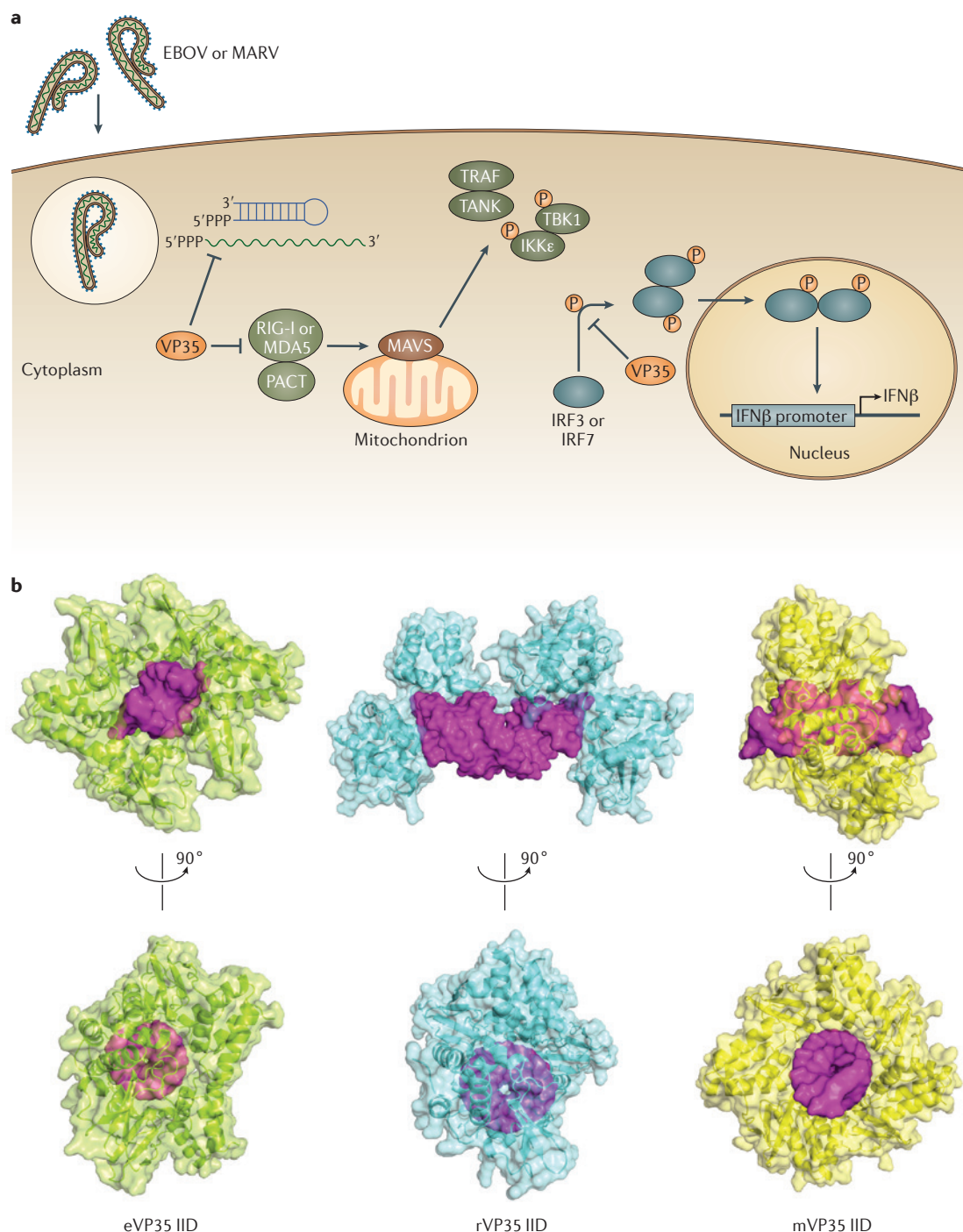


Figure 2 | Subversion of IFN induction. a | Schematic of the interferon (IFN) production pathway and mechanisms of filoviral evasion. The image depicts how Ebola virus (EBOV) viral protein 35 (eVP35) and Marburg virus (MARV) VP35 (mVP35) antagonize signalling pathways that lead to the expression of type I IFNs. RIG-I-like receptors (RLRs), which include RIG-I and melanoma differentiation-associated protein 5 (MDA5), detect RNA products of viral replication, such as cytoplasmic double-stranded RNAs (dsRNAs) or RNAs with 5' triphosphates. Activation of RLRs is facilitated by the protein PACT (a protein activator of the IFN-induced antiviral kinase protein kinase R (PKR)). Upon activation, RLRs signal through the mitochondrial antiviral-signalling protein (MAVS) to activate kinases I κ B kinase- ϵ (IKK ϵ) and TBK1. These kinases phosphorylate IFN regulatory factor 3 (IRF3) or IRF7, which then accumulates in the nucleus and promotes expression of type I IFNs. VP35 can bind to dsRNAs and to PACT, preventing RLR activation. In addition, VP35 interacts with and acts as a decoy substrate for IKK ϵ and TBK1. **b** | Structural analysis of the binding of the IFN inhibitory domain (IID) of filoviral VP35 proteins to dsRNAs. This analysis reveals differences among the dsRNA recognition mode for eVP35 (RCSB Protein Data Bank (PDB) entry 3L25), Reston ebolavirus (RESTV) VP35 (rVP35; PDB entry 4L62) and mVP35 (PDB entry 4GHL). eVP35 is bound to 8 bp dsRNA; rVP35 is bound to 12 bp dsRNA; and mVP35 is bound to 18 bp dsRNA. dsRNA is shown in magenta. TANK, TRAF family member-associated NF- κ B activator; TRAF, TNF receptor-associated factor.

VP35 can bind to dsRNA and suppress RLR-mediated activation of IFN responses induced by Sendai virus infection or by transfected dsRNAs, the ability of VP35 point mutants that lack dsRNA-binding activity to suppress RLR-induced type I IFN production is greatly reduced^{52,92–95}. These data suggest a model in which VP35 sequesters RLR-activating dsRNAs. Supporting this view, purified VP35, but not VP35 mutants that lack dsRNA-binding activity, can prevent dsRNA-mediated activation of RIG-I ATPase *in vitro*⁹⁶.

VP35 was also observed to interact with PACT (also known as PRKRA), a multifunctional cellular protein that was originally described as a protein activator of the IFN-induced antiviral kinase protein kinase R (PKR)⁹⁷. PACT has also been shown to interact with and facilitate the activation of RIG-I⁹⁸; VP35 blocks this interaction, thereby preventing PACT-mediated activation of RIG-I both *in vivo* and *in vitro*. Interestingly, the interaction of VP35 with PACT can occur in the absence of dsRNA but requires the same patch of basic amino acids in the IID that mediates the interaction of VP35 with dsRNA⁹⁶. Therefore, these basic amino acid residues are capable of participating in either protein–dsRNA or protein–protein interactions, as highlighted by the X-ray crystal structure of VP35 complexed with dsRNA (FIG. 2b). VP35 point mutants lacking dsRNA-binding activity failed to interact with PACT and to block PACT-mediated activation of RIG-I. Demonstrating the relevance of PACT to EBOV infection, PACT overexpression overcomes the inhibitory effects of VP35 and triggers an IFN response in cells infected with wild-type EBOV. By contrast, the IFN response induced by EBOV carrying a mutated VP35 is decreased in PACT-knockdown cells⁹⁶. In summary, current models suggest that VP35 can inhibit RLR responses both by the sequestration of immunostimulatory dsRNAs and through interaction with PACT (FIG. 2). These inhibitory mechanisms rely on the VP35 IID and contribute to the potent suppression of the type I IFN response in filovirus-infected cells.

In addition to suppressing the production of type I IFNs, VP35-mediated inhibition of RLR signalling suppresses the maturation and function of dendritic cells. As noted above, dendritic cells are important targets of filovirus infection *in vivo*^{27,49–51}. The role of dendritic cells is to stimulate adaptive immune responses upon encountering a pathogen⁹⁹, which requires dendritic cells to undergo a ‘maturation’ process triggered by innate immune signalling pathways. However, *in vitro* infection of dendritic cells with filoviruses results in aberrant dendritic cell maturation^{49–51}. By contrast, EBOV variants carrying mutant versions of VP35 induce dendritic cell maturation. Furthermore, the expression of eVP35 in dendritic cells recapitulates the suppressive effects of infection with wild-type EBOV, and the inhibitory effects are largely mediated by the ability of VP35 to inhibit RLR activity^{49,51–53,100}. Therefore, the effect of RLR suppression by VP35 probably extends beyond the production of type I IFNs, and also impairs the activation of adaptive immune responses.

Structural determinants of VP35 binding to dsRNA. The dsRNA-binding activity of both eVP35 and mVP35 has been explained using structural and biophysical methods. Overall, VP35 contains two identifiable domains, an amino-terminal oligomerization domain and a C-terminal IID. VP35 residues 220 to 340, which constitute the IID, were defined as the minimal region important for dsRNA binding and IFN inhibition using a combination of limited proteolysis and nuclear magnetic resonance (NMR)-based studies⁹¹. Consistent with this finding, the C-terminal IID is sufficient for suppression of IFN production, but the addition of the oligomerization domain enhances this inhibition¹⁰¹.

The structure of the eVP35 IID, which has an α -helical subdomain and a β -sheet subdomain, contains two regions with basic charge. These regions, termed the first basic patch (FBP) and the central basic patch (CBP), are functionally important: the FBP residues, which are conserved among EBOVs, seem to be important for virus replication, but not for dsRNA binding or IFN inhibition¹⁰², whereas the CBP is involved in dsRNA binding and IFN inhibition^{91,94,103}. Comparison of the IIDs of EBOV, RESTV and MARV reveals remarkable similarities in the overall structure as well as the highly conserved molecular surfaces^{22,23,94,103} (FIG. 2b). Moreover, evaluation of the RESTV and MARV IIDs confirms that the CBP is conserved among filoviruses, but EBOV has the most extensively charged CBP^{94,104}.

Despite the structural and functional similarities described above, the EBOV, MARV and RESTV IIDs also show differences in their biochemical and structural characteristics. For example, *in vitro*, eVP35 can bind to dsRNA fragments that are at least 8 bp in length; VP35 from RESTV (rVP35) can bind to dsRNA fragments that are longer than 12 bp; and mVP35 binds to dsRNA fragments with at least 18 bp²² (FIG. 2b). Furthermore, for dsRNA fragments that are longer than 18 bp, eVP35 binds with greater affinity than mVP35. Data also suggest that eVP35 and mVP35 show significant differences in how they bind to dsRNA. For example, eVP35 displays two different binding modes, recognizing both blunt ends of dsRNA and the dsRNA backbone, whereas mVP35 can only recognize the dsRNA backbone (FIG. 2b). Because the 5' triphosphate end plays a crucial part in RIG-I activation, the ability of eVP35 to mask this motif may provide eVP35 with an increased capacity to block IFN production.

Subversion of IFN-induced signalling

In addition to preventing cytosolic sensing of viruses and induction of an IFN response, EBOV and MARV block the cellular response to type I IFNs. As a consequence, IFN-induced gene expression is substantially impaired in EBOV-infected and MARV-infected cells^{77,78}. In uninfected cells, the addition of type I IFNs activates a JAK–STAT signalling pathway in which receptor-associated JAK family kinases JAK1 and TYK2 are activated. These kinases phosphorylate STAT1 and STAT2, which induces the formation of STAT1 homodimers or STAT1–STAT2 heterodimers¹⁰⁵. The phosphorylation and dimerization of STAT1 reveal a non-conventional

nuclear localization signal (ncNLS) on STAT1 that can interact with any of the three members of the NPI-1 subfamily of karyopherin- α (KPNA) proteins, which comprises KPNA1, KPNA5 and KPNA6 (REF. 106). The interaction of tyrosine-phosphorylated STAT1 (pSTAT1) with KPNAs mediates the import of the STAT protein dimers into the nucleus¹⁰⁶, where these proteins can bind to ISREs in order to regulate transcription (FIG. 3a).

eVP24 interferes with STAT nuclear entry. In EBOV-infected cells, sensing of type I IFNs by IFNAR triggers STAT phosphorylation; however, nuclear entry of pSTAT1 is impaired by eVP24 (REF. 21). eVP24 interacts with the same NPI-1 subfamily of KPNAs that are bound by pSTAT1, and mapping studies suggest that eVP24 occupies the pSTAT1-binding site on KPNA, thereby acting as a competitive inhibitor of pSTAT1 (REF. 107). Structural and biochemical studies support this competition model of eVP24 binding to KPNA. eVP24 forms a pyramid-shaped tertiary structure with a series of α -helices resting on a β -sheet pedestal^{108,109} (FIG. 3b). The KPNA proteins possess an N-terminal importin- β -binding domain, ten armadillo (ARM) repeats (numbered 1–10) and a C-terminal domain. Whereas proteins with classical, basic NLSs (cNLSs) bind to the central ARM repeats of KPNAs (repeats 2–4 or 6–8), the ncNLS of pSTAT1 interacts with C-terminal regions of the NPI-1 KPNAs, including ARM repeats 8–10. A recent study generated a stable truncated KPNA5 construct (termed KPNA5C), which contains ARM repeats 7–10 (REF. 110). The structure of KPNA5C in complex with eVP24 identified three regions within eVP24 — termed clusters 1, 2 and 3 — that interact with KPNA5C ARM repeats 8, 9 and 10, respectively (FIG. 3b). Moreover, residues within KPNA5C identified as crucial for eVP24 binding were also important for STAT1–KPNA binding, directly correlating binding with inhibitory activity. Thus, eVP24 targets well-conserved, functionally significant interactions in KPNA to prevent STAT1 binding to KPNA, inhibiting the host IFN-mediated response¹¹⁰. This study also revealed that the KPNA-binding site of monopartite cNLSs is largely unperturbed by eVP24 binding, suggesting that eVP24 may preserve nuclear import of cellular cargo that relies on these localization signals. However, bipartite cNLSs as well as ncNLSs may be affected by eVP24–KPNA binding^{110,111}. Therefore, additional work will be required to determine whether eVP24 mainly affects nuclear import of pSTAT1 or whether import of other cargo is also affected.

mVP40 blocks JAK1 signalling. In contrast to eVP24, MARV VP24 (mVP24) does not bind to KPNAs or significantly alter IFN-induced gene expression²⁴. The inhibition of IFN signalling in MARV-infected cells is instead mediated by mVP40, which blocks the activation and function of JAK1 (REF. 24) (FIG. 3a). This blockade recapitulates the phenotype of JAK1-knockout cells in that JAK family kinase, and STAT protein tyrosine phosphorylation events were absent after addition of either type I IFNs or IFN γ ¹¹². Overexpression studies also suggest that mVP40 acts by specifically inhibiting

Figure 3 | **Subversion of IFN-induced signalling and of IFN-induced protein tetherin.** **a** | Schematic of the interferon (IFN) response pathway and mechanisms of filoviral evasion. Type I IFNs bind to the extracellular domains of the heterodimeric type I IFN receptor (IFNAR), which is composed of the subunits IFNAR1 and IFNAR2. This activates receptor-associated tyrosine kinases Janus kinase 1 (JAK1) and TYK2. JAK1 and TYK2 phosphorylate signal transducer and activator of transcription 1 (STAT1) and STAT2, which leads to the formation of STAT1–STAT1 homodimers or STAT1–STAT2 heterodimers. Tyrosine-phosphorylated STAT (pSTAT) dimers associate with karyopherin α 1 (KPNA1), KPNA5 and KPNA6, and are transported into the nucleus, where they bind to IFN-sensitive response elements (ISREs). Binding of pSTAT dimers to ISREs induces the expression of IFN-stimulated genes (ISGs), which include the genes that encode the antiviral kinase protein kinase R (PKR) and major histocompatibility complex I (MHC) class I molecules. Marburg virus (MARV) VP40 (mVP40) inhibits the IFN-induced activation of JAK1. Ebola virus (EBOV) VP24 (eVP24) binds to KPNA1, KPNA5 and KPNA6 to block KPNA–STAT1 interactions. Both mechanisms inhibit ISG expression. **b** | eVP24 (RCSB Protein Data Bank (PDB) entry 4M0Q) and Marburg virus (MARV) VP24 (mVP24) (PDB entry 4OR8) show a similar pyramidal structure (left panel)¹¹⁰. eVP24 uses a unique non-classical nuclear localization site to interact with KPNA1, KPNA5 or KPNA6. eVP24 binds to the KPNA5 C terminus (orange) (PDB entry 4U2X) (right panel, superimposed on full length KPNA from *Saccharomyces cerevisiae* (PDB entry 1BK5)). **c** | When the filovirus VP40 protein is overexpressed in the absence of glycoprotein (GP), the IFN-inducible protein tetherin can inhibit budding of virus-like particles (VLPs). **d** | EBOV GP counteracts the antiviral effects of tetherin, enabling budding and release of EBOV. IRF9, IFN regulatory factor 9.

JAK1. Whereas overexpression of JAK1 triggers JAK1 autophosphorylation and STAT1 and STAT2 phosphorylation, mVP40 co-expression prevents these phosphorylation events. By contrast, when TYK2 was overexpressed, phosphorylation of TYK2, STAT1 and STAT2 was induced, and expression of mVP40 had little to no effect on these phosphorylation events²⁴. Although JAK1 function is inhibited by mVP40, the underlying mechanism remains to be defined, and direct interaction of mVP40 with JAK1 has not been demonstrated. Thus, in contrast to EBOV, MARV abrogates IFN signalling by blocking the type I IFN-induced phosphorylation of JAK1, STAT1 and STAT2.

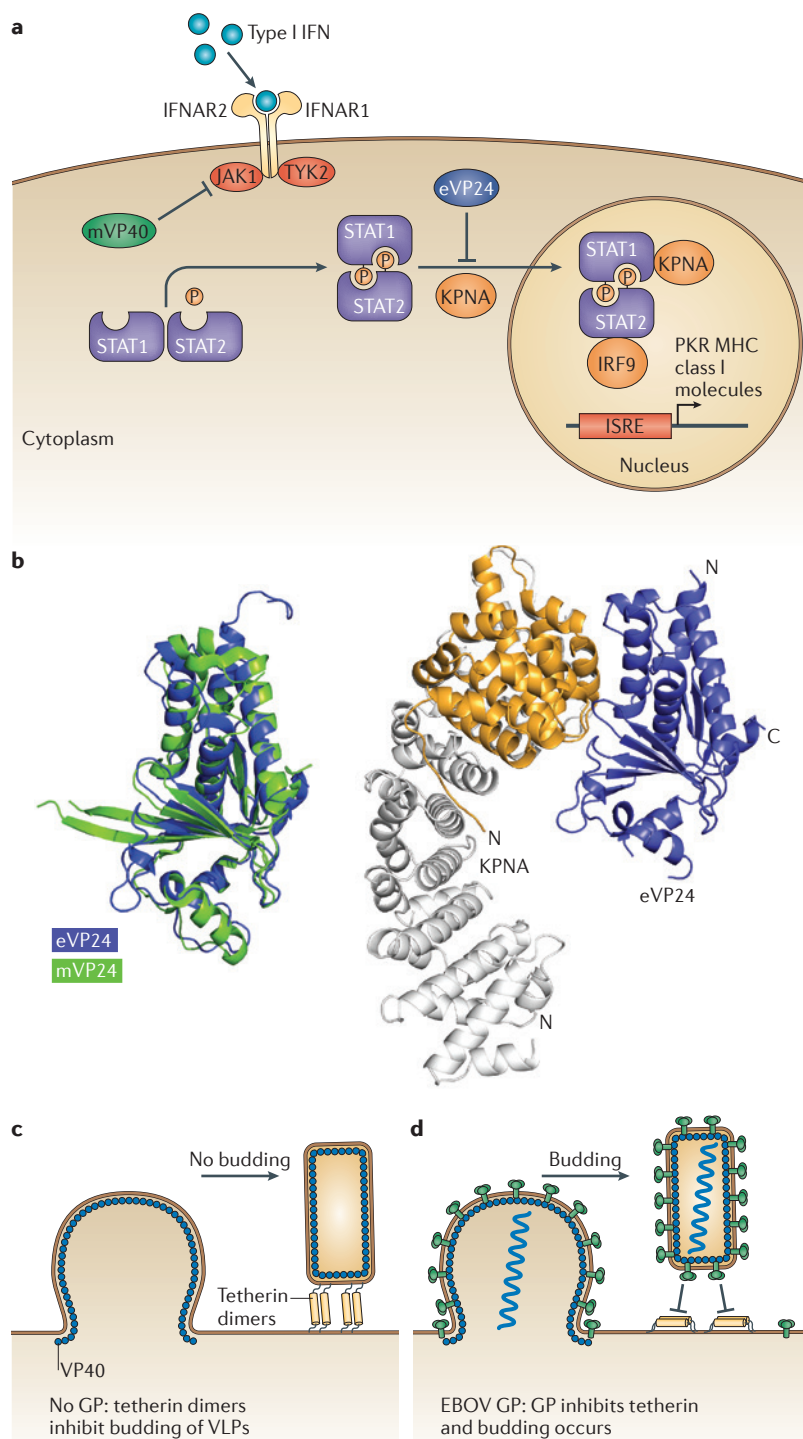
Inhibition of IFN-induced antiviral proteins. In addition to blocking cytosolic sensing and cellular signalling pathways, filoviruses rely on mechanisms that counteract selected IFN-induced antiviral proteins, including PKR and tetherin. PKR is a well-characterized IFN-induced, dsRNA-activated kinase with antiviral activity, and VP35 has been shown to block its activity^{113,114}. Although PKR is activated by dsRNA, some VP35 mutants that are unable to bind to dsRNA retain the capacity to inhibit PKR, suggesting that VP35 inhibits PKR by a dsRNA-independent mechanism of action. Although the mechanism of inhibition remains to be elucidated, it is intriguing

Monopartite

Involving one part. In the case of nuclear localization signals, the recognized sequence is composed of a single stretch of amino acids.

Bipartite

Involving two parts. In the case of nuclear localization signals, the recognized sequence is composed of two separate amino acid sequences connected by a linker region.



that the VP35 interactor PACT is an activator of PKR, suggesting that VP35 may block activation of PKR by PACT^{97,115}. Furthermore, the effect of PKR inhibition on EBOV replication also remains to be defined. The biological significance of this function is suggested by the observation that phosphorylation of PKR and its substrate eukaryotic initiation factor 2 α (eIF2 α) was not detected in EBOV-infected 293 cells, and by the observation that the status of eIF2 α phosphorylation regulates viral replication in cells that are persistently infected with EBOV *in vitro*^{114,116}.

Tetherin is an IFN-inducible protein that inhibits the budding and release of some enveloped viruses by tethering the particles to the cell surface^{117,118}. The release of virus-like particles (VLPs), which occurs upon overexpression of either the EBOV or MARV VP40 matrix proteins, is likewise inhibited by tetherin^{119,120} (FIG. 3c). In contrast to the tetherin-mediated inhibition of budding seen when VP40 was expressed outside the context of virus infection, VLP release from EBOV-infected cells was relatively unaffected by tetherin because filoviral GPs counteract the anti-budding effects of tetherin^{121–123} (FIG. 3d). The molecular basis for GP antagonism of tetherin remains ambiguous, but GP does not alter tetherin expression, decrease tetherin levels at the cell surface or prevent tetherin association with lipid rafts, which are enriched for tetherin^{123–125}.

Subversion of other host pathways. There is accumulating evidence that filoviruses not only block innate immune responses but that they also adapt to their host in order to sustain viral replication and to subvert the translational regulation mechanisms of host cells. For example, cellular stress can lead to a general decrease in translation through the phosphorylation of the translation initiation factor eIF2 α (FIG. 4). This suppression of translation can serve as a mechanism that allows cells to recover from stress conditions or, in the case of viral infection, can serve as a means to impair translation of viral proteins, affecting virus replication. However, translation of some cellular mRNAs that contain upstream ORFs (uORFs) is increased under these conditions¹²⁶, and EBOV seems to mimic this strategy as a mechanism of subverting the host cell.

Filoviruses are unusual in comparison to many other non-segmented, negative-strand RNA viruses in that they produce mRNAs with long 5' and 3' untranslated regions (UTRs). The 5' UTRs of the EBOV mRNAs encoding VP35, VP30, VP24 and L contain upstream AUGs (uAUGs), and the uAUGs of the EBOV mRNAs encoding VP30, VP24 and L mark the beginning of short uORFs¹²⁷. When assessed in reporter gene assays under normal conditions, the uORF in the L-encoding mRNA was found to suppress expression of the primary ORF of L by approximately tenfold. This suppression required the uAUG and reflects decreased translation initiation at the AUG of the primary ORF (FIG. 4a). However, in cells that were stressed by addition of thapsigargin, which triggers phosphorylation of eIF2 α , the uORF can sustain translation of the L ORF (the primary ORF) by increasing translation initiation efficiency at the primary AUG (FIG. 4b). This suggests a model whereby an uORF helps EBOV to maintain expression of its L polymerase under conditions of stress and eIF2 α phosphorylation, such as those that may occur when the innate immune response leads to PKR activation (FIG. 4b). Furthermore, replication in a virus mutant lacking the L gene uAUG site was impaired by approximately tenfold relative to replication of wild-type virus and showed increased sensitivity to stress induced by thapsigargin, demonstrating that the L uORF is important for EBOV replication¹²⁷.

293 cells
Human embryonic
kidney-derived cell line.

Lipid rafts
Microdomains within a lipid membrane. It is postulated that lipid rafts support the concentration of viral proteins for more efficient viral assembly and budding.

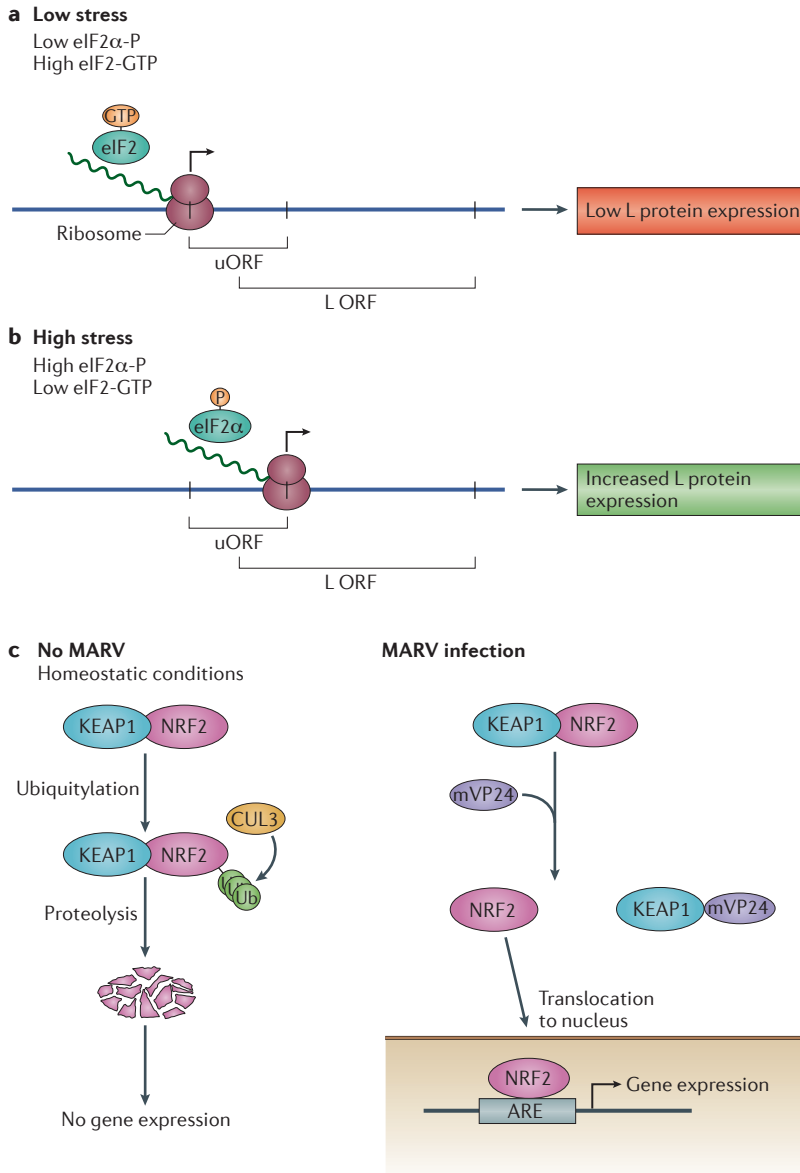


Figure 4 | Subversion of host translation mechanisms to support virus replication.

a | It has been observed that several of the 5'-capped viral mRNAs from Ebola virus (EBOV) possess upstream ORFs (uORFs) in their 5' untranslated regions (UTRs). Under low-stress conditions, when levels of phosphorylated eukaryotic initiation factor 2α (eIF2α-P) are low and levels of eIF2-GTP are high, the uORF present in the mRNA encoding the large protein (L) attenuates L translation initiation. This is because scanning ribosomes initiate translation efficiently at the start AUG codon present in the uORF, rather than at the primary AUG start codon. **b** | Under conditions of stress, such as stress owing to activation of innate immune responses, eIF2α phosphorylation increases, leading to increased levels of eIF2α-P, and translation initiation at the uORF becomes less efficient. This allows the scanning ribosome to bypass the AUG start codon located in the uORF and increases initiation at the primary AUG, helping to sustain L expression despite the increase in eIF2α phosphorylation. **c** | Under homeostatic conditions, kelch-like ECH-associated protein 1 (KEAP1) interacts with transcription factor nuclear factor (erythroid-derived 2)-like 2 (NRF2) and promotes its cullin 3 (CUL3)-dependent ubiquitylation and proteasomal degradation, thereby suppressing expression of antioxidant response genes. Marburg virus (MARV) viral protein 24 (mVP24) interacts with KEAP1, preventing its interaction with NRF2. This allows NRF2 to accumulate in the nucleus and to activate expression of genes with antioxidant response elements (AREs). The antioxidant response genes encode proteins with cytoprotective effects. This may preserve the life of MARV-infected cells to enhance virus replication.

In another example, mVP24 has been demonstrated to interact with and modulate the function of cellular protein kelch-like ECH-associated protein 1 (KEAP1)^{109,128,129}. KEAP1 normally directs cullin 3 (CUL3) ubiquitin ligase to target proteins such as nuclear factor (erythroid-derived 2)-like 2 (NRF2), a transcription factor that binds to and activates promoters possessing antioxidant response element (ARE) sequences. ARE-containing genes encode proteins that allow cells to recover from oxidative stress and other stresses. Under homeostatic conditions, NRF2 is constitutively targeted for degradation owing to ubiquitylation directed by KEAP1 (FIG. 4c). However, when a cell experiences stress, the KEAP1–NRF2 interaction can be disrupted, stabilizing NRF2 and promoting the expression of ARE-containing genes¹³⁰. mVP24 interacts with the kelch domain of KEAP1 through an acidic motif present within a loop (called the 'K loop'). This loop possesses the amino acid residues glycine and glutamic acid preceded by acidic residues, which is reminiscent of interacting motifs on other KEAP1-binding partners, including NRF2. The specificity of mVP24 for the KEAP1 kelch domain is derived, in large part, from the extended β-strands that protrude from mVP24 and presumably that can reach into the deep binding pocket within the kelch structure^{109,131}. Interestingly, eVP24 lacks this structural feature, lacks the K loop and does not bind to KEAP1. Functionally, interaction between mVP24 and KEAP1 disrupts the KEAP1–NRF2 interaction and activates the expression of ARE-containing genes¹⁰⁹ (FIG. 4c). Consistent with the binding data, eVP24 did not upregulate the expression of such genes, but chimeric eVP24 variants possessing the K loop from mVP24 did induce upregulation of the expression of ARE-containing genes. Furthermore, gene expression was upregulated during MARV infection, but not EBOV infection, of macrophage-like THP1 cells, which is consistent with the observation that mVP24 interacts with KEAP1 but eVP24 does not. These data suggest that turning on antioxidant responses could prolong the survival of virus-infected cells.

Effect of immune evasion on pathogenesis

Among the described filoviral immune evasion functions, those most directly linked to pathogenesis are the ones that directly block innate immune signalling pathways. Recombinant EBOVs possessing mutated VP35 proteins that are unable to bind to dsRNA or PACT trigger nuclear accumulation of IRF3 and induce robust type I IFN responses, whereas wild-type EBOV does not. *In vivo*, these mutant viruses exhibited reduced replication and were non-lethal at the doses tested, providing compelling evidence that the IFN-antagonist function of VP35 is a crucial determinant of EVD *in vivo*^{80,81}.

Studies also implicate the IFN signalling inhibitors eVP24 and mVP40 as virulence determinants in rodent models. Although EBOV does not display virulence in mice or guinea pigs, the virus can be made lethal by serial passage in these animals^{10,11}. The adapted viruses acquire amino acid changes in several genes, including the gene encoding eVP24 (REFS 10,11). In the guinea pig-adapted EBOV, changes in the gene encoding eVP24

Kelch

A repeated motif within proteins that predicts a conserved β -propeller structure.

were sufficient to confer lethality¹³² and, for mouse-adapted EBOV, a mutation in the gene encoding eVP24 combined with a mutation in the gene encoding NP made the virus lethal to mice¹³³. It is presently unclear to what extent the adaptive changes are related to eVP24 IFN-antagonist activities, although the mouse-adapted EBOV was more resistant to the antiviral effects of IFNs following infection of mouse macrophages than the parental, non-adapted virus¹³³.

Like EBOV, MARV does not cause lethal illness in guinea pigs or mice unless it is experimentally adapted to these species by serial passage^{14,134}. Adaptation of the Ravn strain of MARV (RAVV) to mice led to changes across the viral genome; among these were several amino acid changes in mVP40 (REF. 135). When the inhibitory activities of the adapted and parental mVP40 variants towards IFN signalling were compared, the parental mVP40 was impaired in inhibiting IFN responses in mouse cell lines compared with in human cell lines. By contrast, the mouse-adapted mVP40 effectively inhibited IFN responses in both human and mouse cell lines^{136,137}. Based on these data, it seems that the inhibitory function of mVP40 is connected to virulence in mice^{136,137}. Interestingly, the mouse-adapted RAVV virus exhibits budding defects in human cells, and this defect was mapped to the same residues in mVP40 that are implicated in enhanced inhibition of the mouse IFN response. This budding defect correlates with an increased sensitivity to restriction by human, but not mouse, tetherin¹³⁸. Therefore, mVP40 adaptation that results in increased IFN antagonism in mice seems to be associated with functional costs, such as decreased efficiency in viral budding.

Outlook

The mechanisms by which EBOV and MARV counteract host defences are becoming increasingly clear. The VP35 protein from both EBOV and MARV antagonizes several aspects of host defence by blocking the cytoplasmic sensors RIG-I and MDA5, by inhibiting the antiviral kinase PKR and by impairing dendritic cell maturation. In addition to interfering with cytosolic sensing, both viruses are able to antagonize cellular responses to IFNs: the eVP24 protein selectively blocks the nuclear import of pSTAT1, whereas the mVP40 protein counters IFN signalling through inhibition of JAK1. Furthermore, the filovirus GPs counter the antiviral effects of tetherin. In addition, mechanisms whereby EBOV and MARV

capitalize on specific cellular functions, such as stress-regulated translation or antioxidant responses, to promote virus replication are coming into focus. There is also strong evidence that these functions contribute to the severe infections that make filoviruses notable.

Despite this progress in elucidating the mechanisms of immune evasion by filoviruses, fundamental questions remain. For example, it is unclear how these functions relate to specific mechanisms that drive disease, such as inflammation, vascular leakage and coagulopathy, and how they relate to the relative virulence of different filovirus species. Differences in the efficiency of innate immune antagonism, which might arise owing to differences in how viral proteins function and/or in differences in the expression levels of these proteins, could contribute to different degrees of virulence seen between various filoviruses, but such studies are yet to be carried out. Because non-human primate models of filovirus infection are considered to be the gold standard for pathogenesis studies, it will be important to define the contribution of innate immune antagonism and other proposed virulence determinants in these models. Given that filoviruses are zoonotic pathogens that are apparently not pathogenic in their presumed reservoir hosts, it will also be important to understand how filoviral immune evasion functions differ in bats versus humans. If the responses of different hosts diverge, it will be important to understand the mechanistic basis for such differences. Finally, it remains to be determined whether these findings can be exploited for therapeutic development. The West Africa EBOV epidemic has reinforced the need for effective medical interventions, but approved therapeutics are lacking. Given the crucial role of VP35 in virulence, drugs that either disrupt or bypass the inhibitory effects of VP35 on cytosolic sensing and restore RLR signalling in infected cells would be expected to have therapeutic benefit. Similarly, drugs that counteract eVP24 or mVP40 IFN-antagonist functions might enhance the benefit of naturally produced IFNs or synergize with IFN treatment. Studies to identify viral inhibitors of other cellular functions, such as the mechanisms regulating L translation and antioxidant responses, are also needed, as these would be useful as molecular probes for these functions and as possible leads for antiviral development. The need for better therapeutics and for improved understanding of filoviral disease should prompt continued delineation of filovirus innate immune evasion functions.

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Competing interests statement

The authors declare no competing interests.

FURTHER INFORMATION

CDC: <http://www.cdc.gov/>

Outbreaks Chronology: Ebola Virus Disease: <http://www.cdc.gov/vhf/ebola/outbreaks/history/chronology.html>

DATABASES

PDB: <http://www.rcsb.org/pdb/home/home.do>
3L25 | 4L62 | 4GHL | 4M00 | 4OR8 | 4UZX | 1BK5

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